

HDOC 12

Activity Description (Batches 15 & 16):

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases. Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Count the number of even digits, and odd digits. E.g. Number: 12045, Even_dig = 3, Odd_dig = 2
- Find the sum of its even digits and the sum of its odd digits separately. E.g. Number: 12345, Sum_even = 6, Sum_odd = 9

Name: _____

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Batch: _____

1. Algorithm

1. Start

2. Declare variables for choice, num, temp, rem, even_count, odd_count, even_sum, odd_sum.
3. Display the menu options to the user.
4. Read the user's choice.
5. If choice is 1 or 2, prompt the user to input a positive integer num. Validate that $\text{num} > 0$.
6. Use a switch statement based on the value of choice.

7. Case 1 (Count even and odd digits):

- Initialize $\text{even_count} = 0$ and $\text{odd_count} = 0$.
- Set $\text{temp} = \text{num}$.
- Loop while ($\text{temp} > 0$):
 - Calculate $\text{rem} = \text{temp} \% 10$.
 - If $\text{rem} \% 2 == 0$, increment even_count .
 - Else, increment odd_count .
 - Update $\text{temp} = \text{temp} / 10$.
- Display even_count and odd_count , then break.

8. Case 2 (Sum of even and odd digits separately):

- Initialize $\text{even_sum} = 0$ and $\text{odd_sum} = 0$.
- Set $\text{temp} = \text{num}$.

- Loop while (temp > 0):
 - Calculate rem = temp % 10.
 - If rem % 2 == 0, add rem to even_sum.
 - Else, add rem to odd_sum.
 - Update temp = temp / 10.
- Display even_sum and odd_sum, then break.

9. **Default Case:** Display an "Invalid choice" error message.

10. **Stop**

2. Test Case Table

TC ID	Scenario Description	Inputs (choice, num)	Expected Output
TC-01	Valid inputs for Menu Option 1	Choice: 1 Num: 12045	Even_dig = 3 Odd_dig = 2
TC-02	Valid inputs for Menu Option 2	Choice: 2 Num: 12345	Sum_even = 6 Sum_odd = 9
TC-03	Number with all even digits (Option 1)	Choice: 1 Num: 2468	Even_dig = 4 Odd_dig = 0
TC-04	Number with all odd digits (Option 2)	Choice: 2 Num: 1357	Sum_even = 0 Sum_odd = 16
TC-05	Invalid negative number input	Choice: 1 Num: -250	Error: Please enter a strictly positive integer.
TC-06	Invalid menu choice	Choice: 5	Invalid choice. Please choose 1 or 2.

3. C Program Source Code

```
#include <stdio.h>

int main() {
    int choice;
    long long num, temp;
    int rem;
```

```

int even_count = 0, odd_count = 0;
int even_sum = 0, odd_sum = 0;

// Display Menu
printf("--- Digit Operations Menu ---\n");
printf("1. Count the number of even and odd digits\n");
printf("2. Find the sum of even digits and odd digits separately\n");
printf("Enter your choice (1 or 2): ");
scanf("%d", &choice);

// Validate Choice and Take Number Input
if (choice == 1 || choice == 2) {
    printf("Enter a positive integer: ");
    scanf("%lld", &num);

    if (num <= 0) {
        printf("Error: Please enter a strictly positive integer.\n");
        return 1; // Exit program due to invalid input
    }
}

temp = num;

// Switch Case Logic
switch (choice) {
    case 1:
        // Count even and odd digits
        while (temp > 0) {
            rem = temp % 10;
            if (rem % 2 == 0) {
                even_count++;
            } else {
                odd_count++;
            }
            temp /= 10;
        }
        printf("\nNumber: %lld\n", num);
        printf("Even_dig = %d\n", even_count);
        printf("Odd_dig = %d\n", odd_count);
        break;

    case 2:
        // Sum of even and odd digits
        while (temp > 0) {
            rem = temp % 10;
            if (rem % 2 == 0) {
                even_sum += rem;
            }
        }

```

```
        } else {
            odd_sum += rem;
        }
        temp /= 10;
    }
    printf("\nNumber: %lld\n", num);
    printf("Sum_even = %d\n", even_sum);
    printf("Sum_odd = %d\n", odd_sum);
    break;

default:
    printf("\nInvalid choice. Please choose 1 or 2.\n");
    break;
}

return 0;
}
```